



INSTITUTE OF
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CHEMISTRY

Challenges in Interpreting Ultrafast Spectroscopy Data: A Case for Machine Learning in Materials Research

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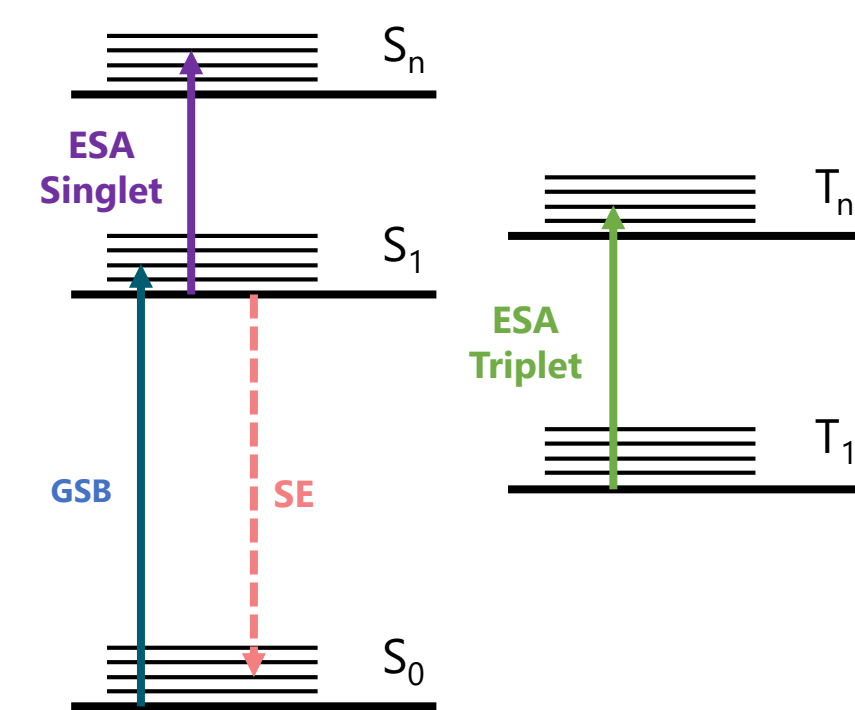
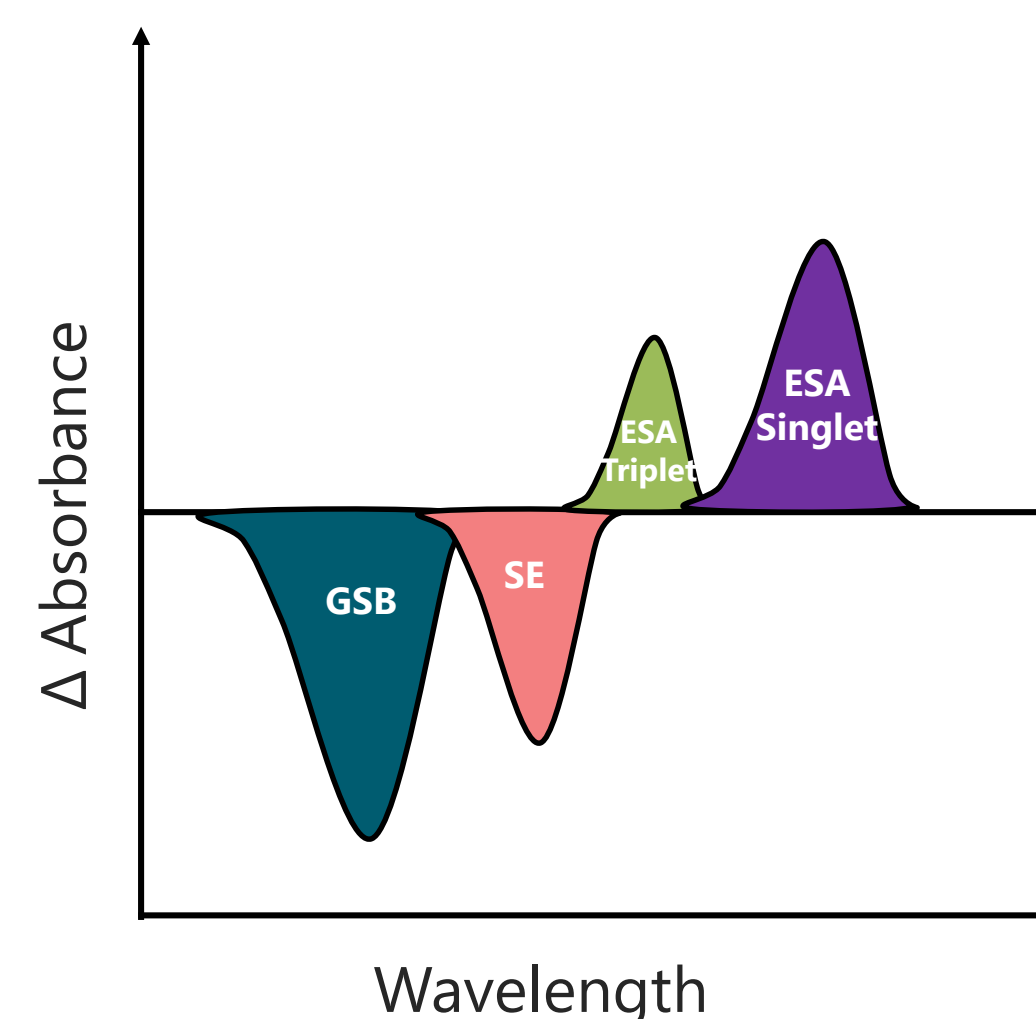
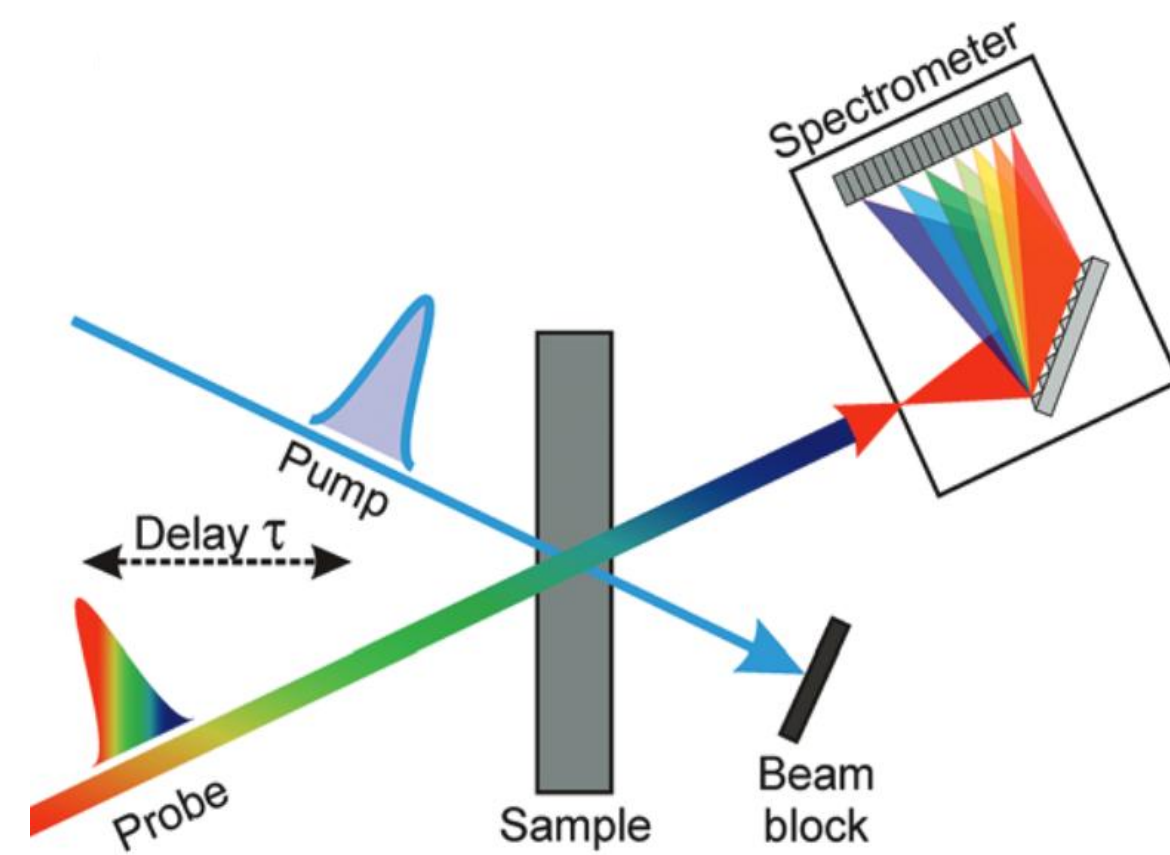
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Introduction

- Transient Absorption (TA) spectroscopy probes photo-induced dynamics on femtosecond-nanosecond timescales.
- **Uncovering Mechanisms:** It is considered one of the most powerful methods for elucidating the mechanisms of excited state dynamics
- **Determining Kinetic Models:** A primary use of the technique is to determine the underlying kinetic model, which explains exactly how a molecular system evolves, relaxes, or transitions between states after absorbing light
- **Driving Technological Innovation:** Due to its wide applicability, it is a crucial tool for characterizing and developing new materials for fields such as organic electronics, solar cell applications, data storage, medicine, etc.

Experiment - transient optical absorption spectroscopy



Pump-probe experimental arrangement.

Spectral features observed in transient absorption spectroscopy.

Classical approach

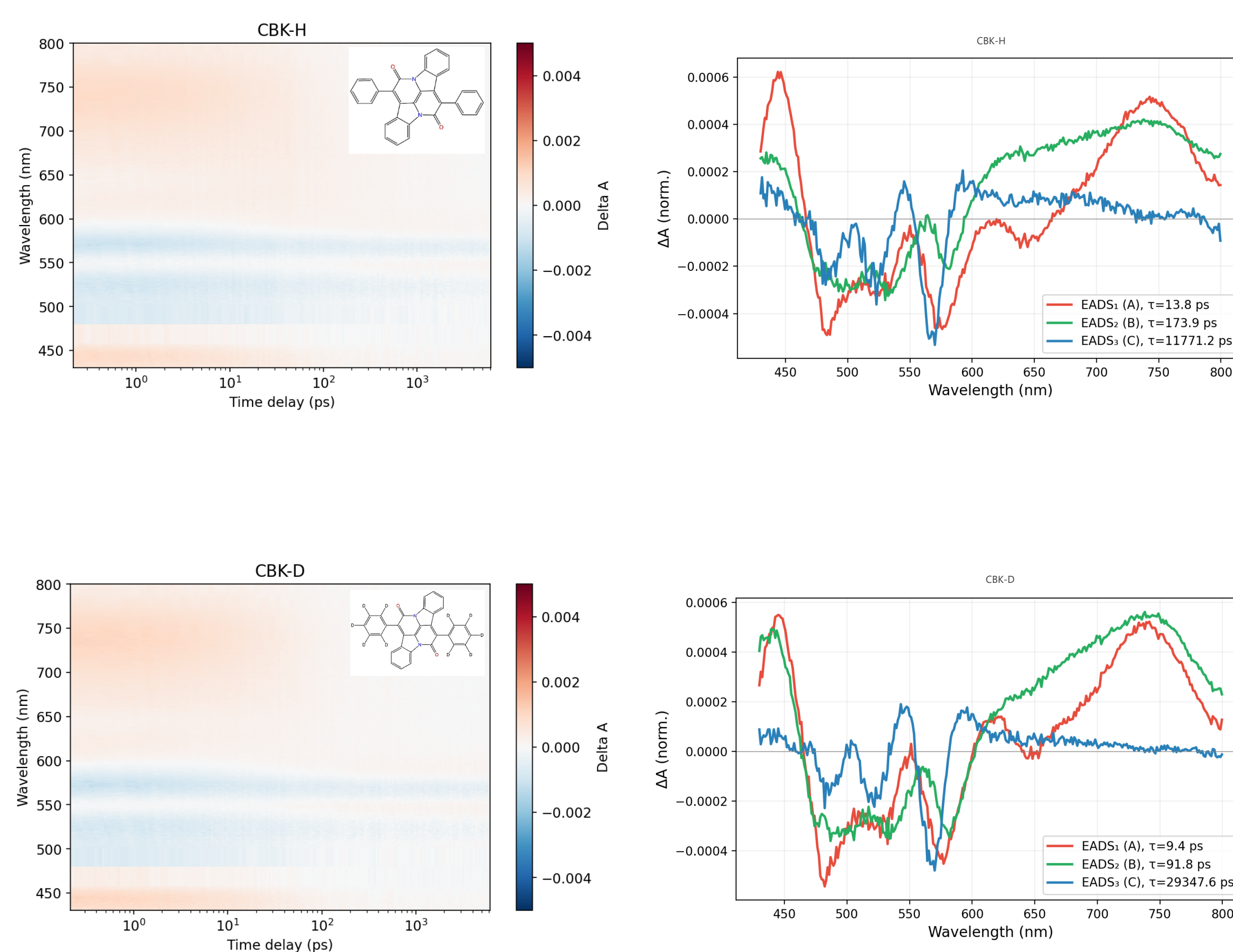
Ultrafast spectroscopy generates massive, complex 2D datasets.

Key Challenges:

- Model ambiguity: different kinetic models fit the same data
- Overlapping, broad excited-state spectral features
- Expert bottleneck: manual model guessing takes hours
- Subjective — different researchers may choose different models

Molecular system under study :

- H/D-substituted indolophthyrindines complex (CBK-H - Protonated, CBK-D - Deuterated).
- Measure the Kinetics by comparing Protonated and Deuterated
- Excitation 476 nm, white-light continuum probe (430–800 nm).
- **Sequential relaxation: A → B → C → Ground State**



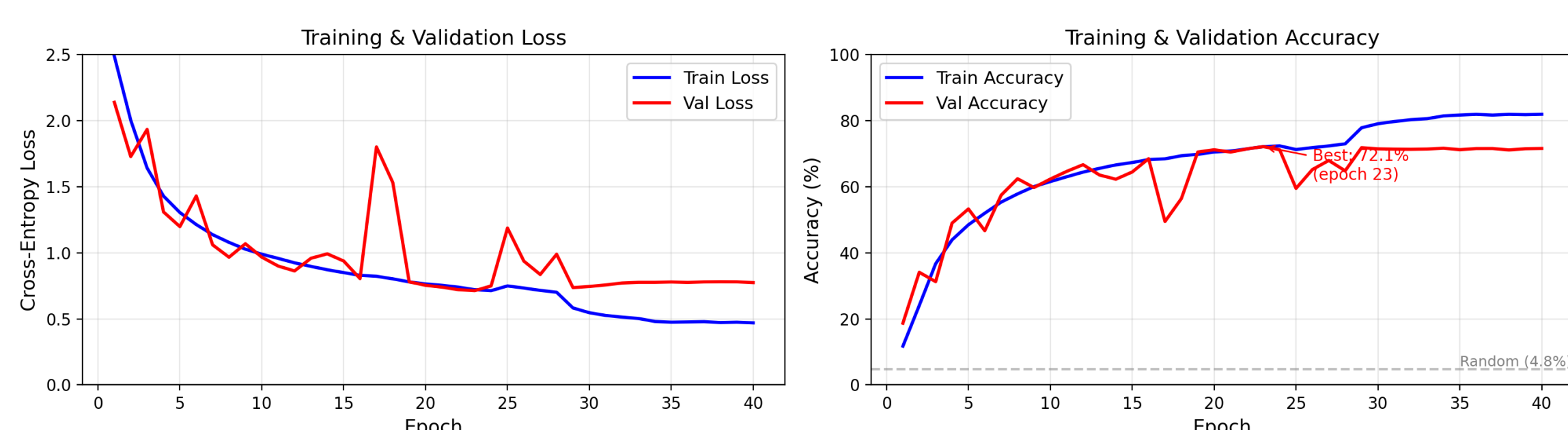
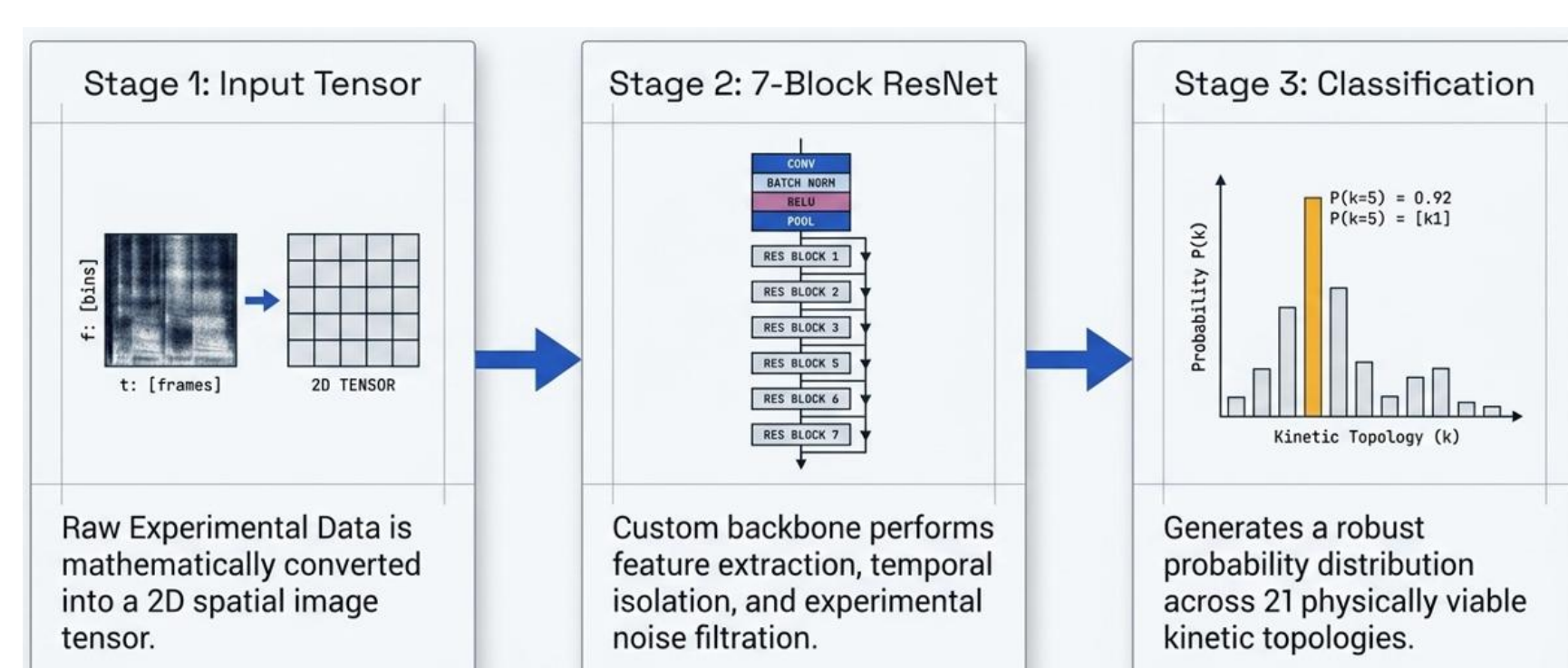
- Global analysis - Model: A → B → C → GS (3 exponentials + Gaussian IRF)

Component	CBK-H	CBK-D
τ_1 (A→B)	13.8 ps	9.9 ps
τ_2 (B→C)	173.9 ps	91.8 ps
τ_3 (C→GS)	>6 ns	>6 ns

- Anomalous effect: τ_1 , τ_2 : excited state of Deuterium is decaying faster.
- Expected: Usually, deuterated organic molecules relax more slowly in the excited state than the normal hydrogen analogs
- Traditional Global Target Analysis relies on assuming a predefined kinetic model with an exponential decays.
- **The Problem: How do we confirm this 3-state exponential model is chemically accurate in both case?**

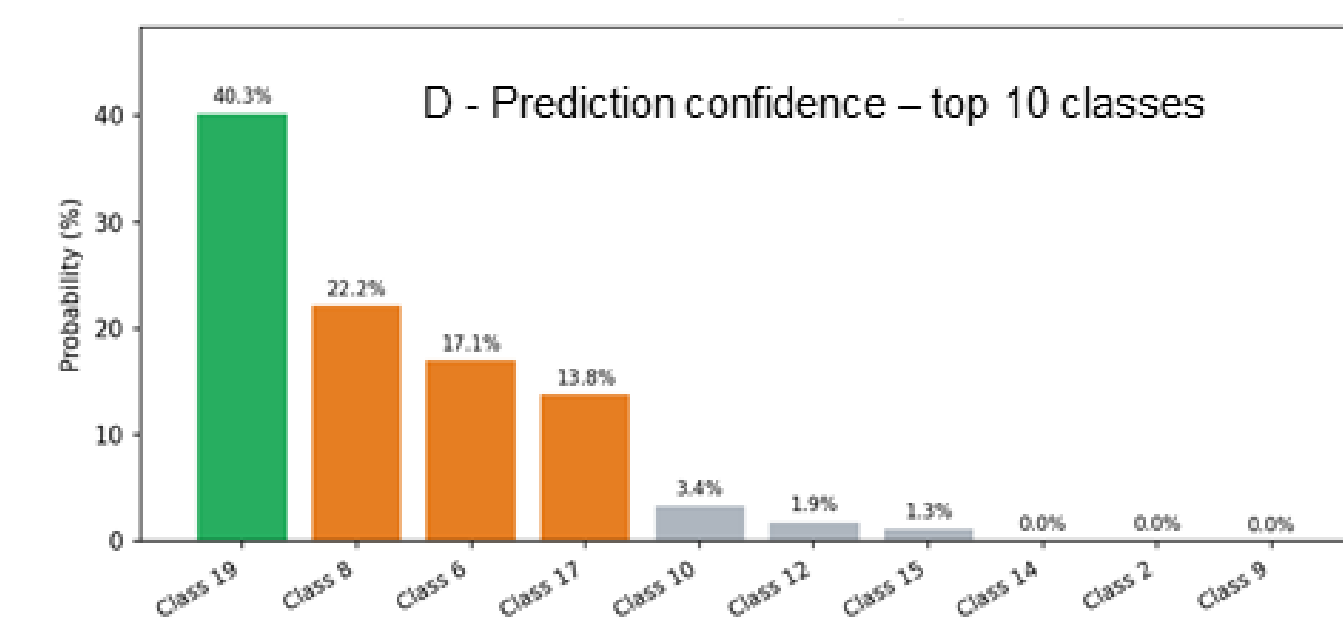
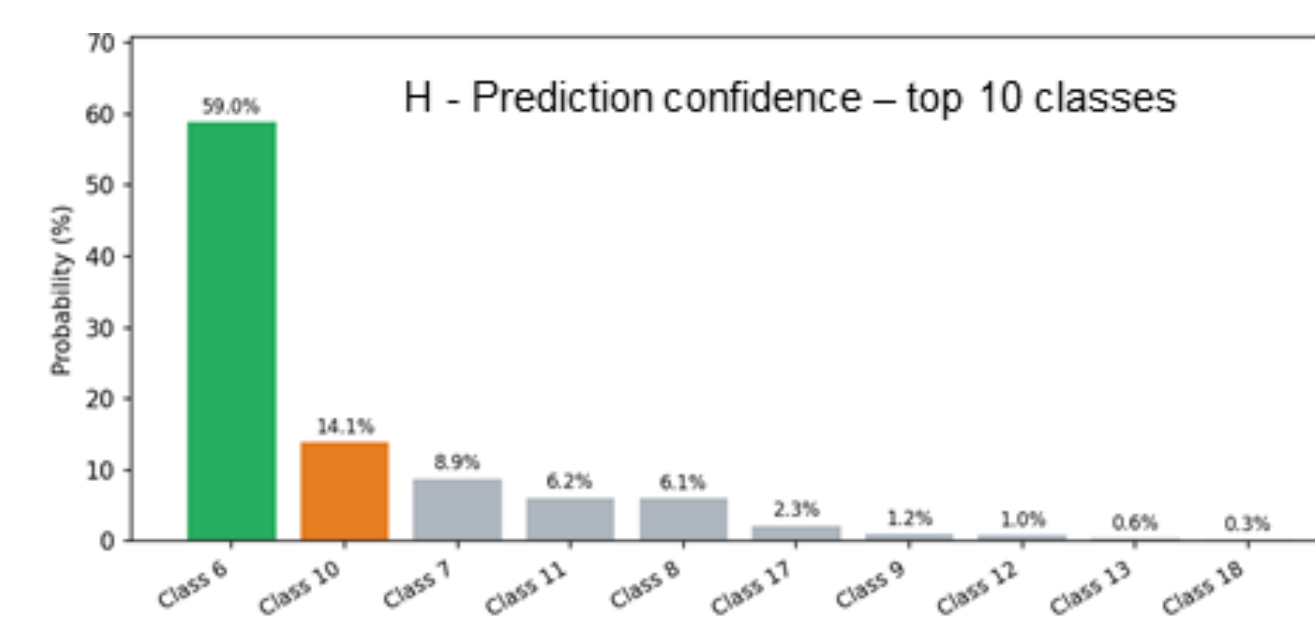
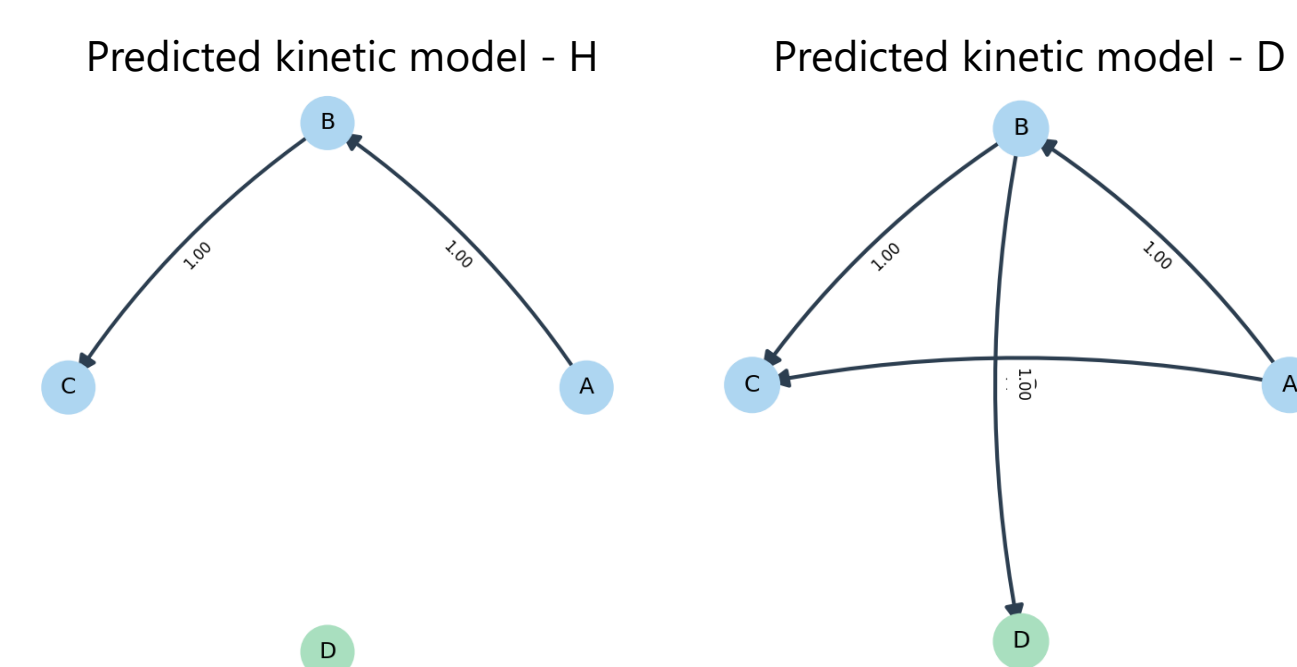
ML APPROACH: DeepSKAN

- ⚙ The Inspiration: We turned to DeepSKAN (Kollenz 2020) for unbiased topological discovery. DeepSKAN (Kollenz et al., J. Phys. Chem. B, 2020) DOI: 10.1021/acs.jpcc.0c04299
- ⚙ **Image-Based Analysis:** Treats raw 2D transient absorption heatmaps directly as images (Time vs. Wavelength).
- ⚙ **Custom 7-Block ResNet:** Uses asymmetric 1D convolutional kernels to isolate temporal decays, then square 2D kernels to extract spatio-temporal features.
- ⚙ **Robust Architecture:** Upgraded with Adaptive Pooling to accept any spectrometer resolution, and Batch Normalization/Dropout to stabilize against real-world noise.
- ⚙ **Artifact-Aware Training:** Trained on a balanced dataset of 50,000 synthetic spectra injected with simulated experimental artifacts (baseline drifts, scatter spikes) to prevent the network from memorizing noise.
- ⚙ **Unbiased Output:** Bypasses human curve-fitting by calculating the mathematical probability across 21 viable kinetic models (up to 4 molecular states).



Results

- ⚙ CBK-H (Protonated): When fed the CBK-H data, DeepSKAN predicted Class 6 with a dominant 59.0% confidence.
- ⚙ Resulting Topology: A pure sequential decay (A→B → C), exactly matching the simple sequential model.
- ⚙ CBK-D (Deuterated): When fed the CBK-D data, the network detected completely different visual patterns. The highest Prediction was Class 19 (40.3% confidence), with other branched models (Class 8, Class 17) making up the rest. Resulting Topology:
- ⚙ A complex branched pathway where state B splits into multiple decay routes.



Conclusion

- ⚙ DeepSKAN predicts that Deuteration might not simply slow down the reaction; it possibly alters the topological mechanism from sequential to branched.
- ⚙ The 'Anomalous behavior' observed in the traditional analysis is likely an artifact of forcing an incorrect model.
- ⚙ It provides a possible decay pathway which can be explored for a better understanding of excited state dynamics

Future Directions & Why I Am Here:

- The "Vibe-Coding" Confession: As a spectroscopy major, I used AI to help me modernize and code this custom PyTorch architecture. It appeared to solved the problem, but I am largely "vibe-coding" the deep learning aspects!
- Looking for Feedback: I came to this Summer School specifically to find out: Am I doing this right? I understand the physical chemistry perfectly, but I am looking for experts to help me understand exactly why my code works so well.
- Call for Collaboration: Seeking ML/DL collaborators to help transition this from an AI-assisted experiment into a rigorously optimized, autonomous pipeline.